Wednesday, March 25, 2026 9:46 AM

2x2x2
We use terraform one ops
2 deployed in openstack
2 deployment happen on azure
2 deployment happen on gcp
No recording
Step by step documentation
30% theoretical 70% hand on

IAC

Wednesday, March 25, 2026 9:59 AM

What is the need of Infra structure as code?

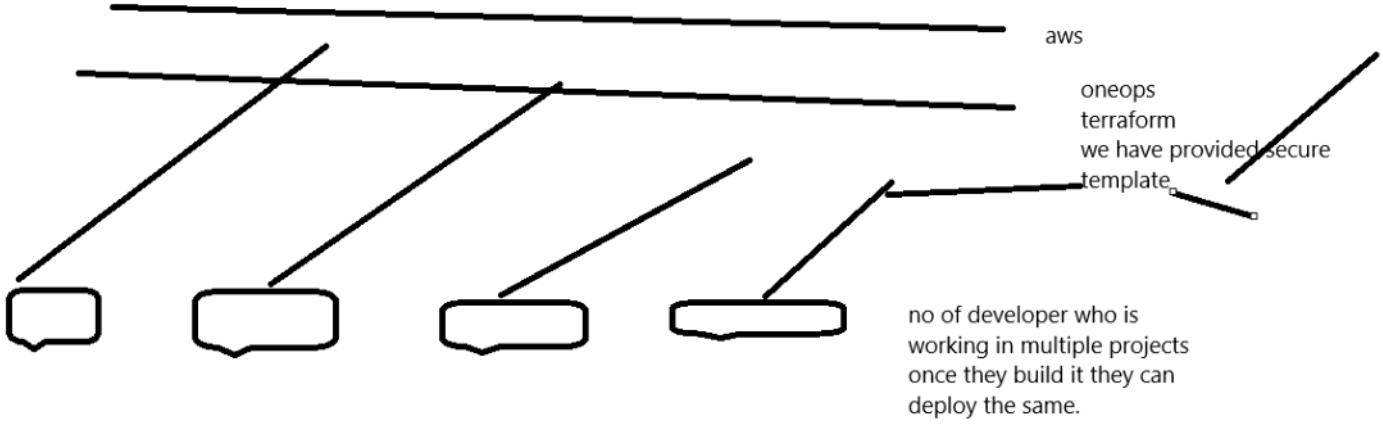
1. Easy deployment
2. Automation
3. Time reduction

What is the biggest challenge in the cloud.
Security: is always an concern in the cloud.

How to avoid this security risk.

Walmart we have more than 10000 application running.

Once you build your app you need to deploy faster
But with security in mind.
We can use iac to build an secure template.



3/25/2026 10:08 AM - Screen Clipping

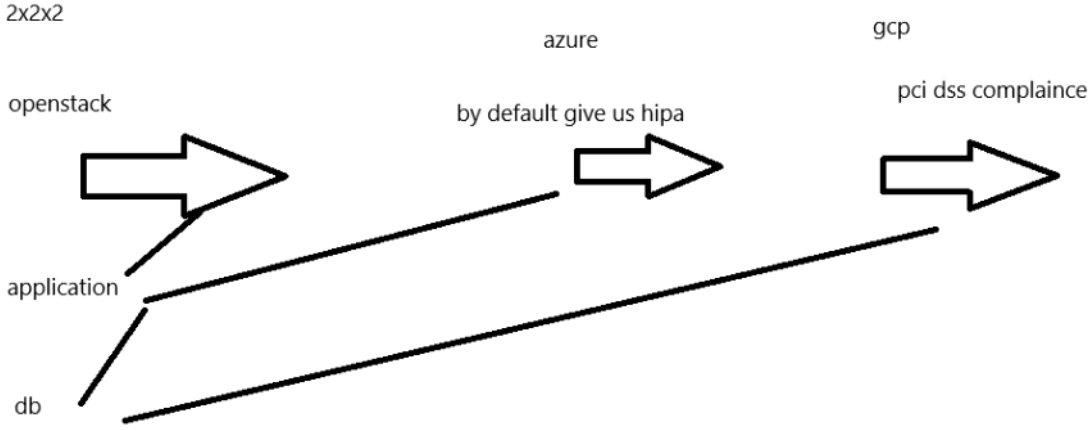
1. We got an secure template to create resources in the cloud.
2. One common template used by everyone.
3. Human mistake
4. Dev test stagin prod
5. Multi cloud deployment

Every cloud service provider has there own native language

1. Aws use cft
2. Azure arm template

Terraform is an single langauge we can use to build the environment. In multi cloud.

HA cloud goes down



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IAC

Wednesday, March 25, 2026 10:22 AM

Infrastructure as code

1. Configuration management tool: on my existing resource we are making changes updating the environment patching ansible chef puppet
2. Provisioning tool: we are building the environment in the cloud terraform comes into provisioing.

Stability

Wednesday, March 25, 2026 10:28 AM

Using terraform we created a well secure template which contains vpc ec2 storage and sg. And build the environment.
Someone login inside the console and add a tag
Next time when I try to do terraform apply automatically the tag which is added will be deleted.

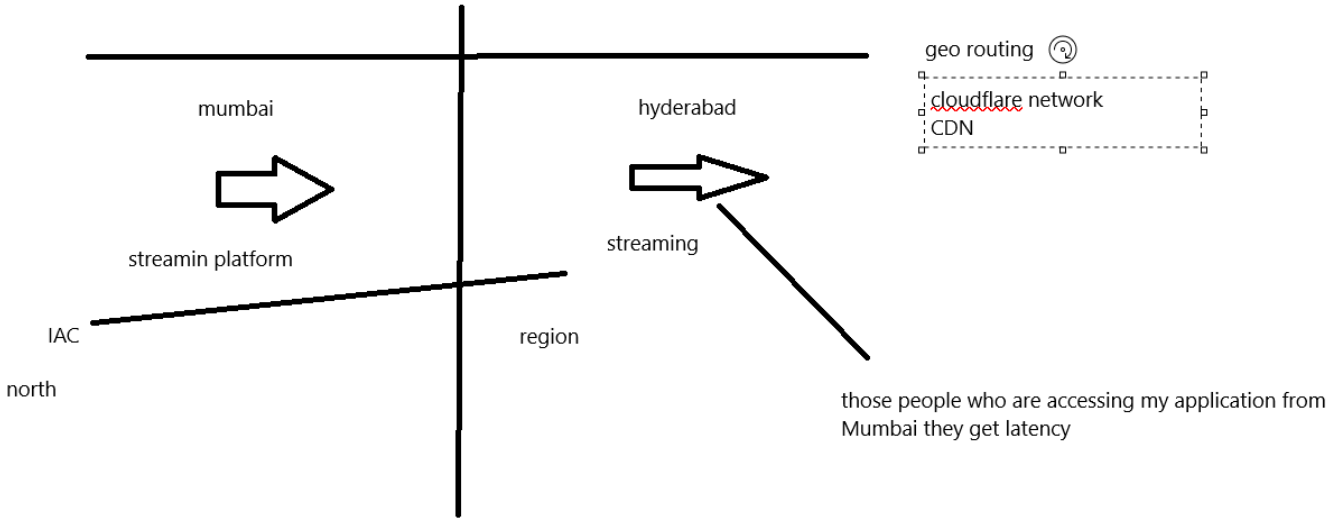
Make changes on those resource this need to be done using terraform only.
Single source of truth will be terraform only

Ec2 instance I am running a vm with 250GB of hdd
But there is an admin who find that 250Gb of hdd is not ok and increase the size to 500GB.
Terraform apply
Come back to 250 Gb empty hdd

So we build a pipeline any changes always required approval.

Drift in terraform

We are running multiple resource in aws
There is a team finance team using youtube udemy reeals and they understand that if they want a proper budgeting we can put tags on the resources . Aws console to add those tags
You also run your terraform script all those tags vanish
Add those tags
Ignore_changes tags



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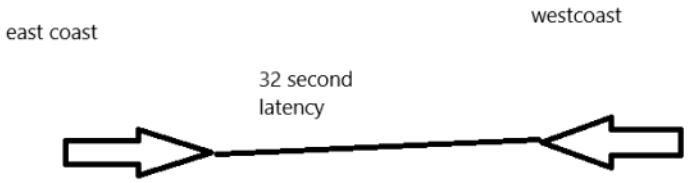

modules in terraform

the main purpose of this training is that how to reuse to code varaibles

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Drift in terraform

They should use my template in dev test environment
Tags
They can open some ports
8080
8081

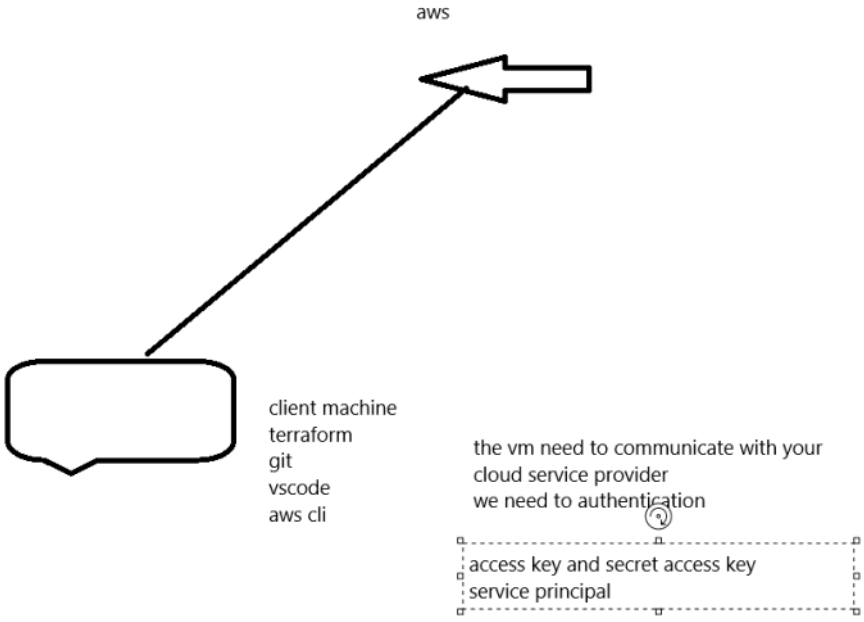
32000 -34000 open this ports using terraform you have opened it all this information will be stored inside your state file
The customer comes to you and tell your that you have opened this port due to which valnurability has come to my environment.
Get the state file
Check your application side
SSD to premium ssd date and timestamp

Demo

Wednesday, March 25, 2026 10:58 AM

We will launch an windows macine
Inside that we will install terraform git vs code.

There is utility
Windows has come with chocolatey package manager
Yum apt command line



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Terraform workflow

Wednesday, March 25, 2026 12:46 PM

Blocks in terraform

Terraform is build using go lang
Terraform has a wrapper language. Which is simple english.
All the devops tools
Ansible chef puppet docker k8s terraform
Already the documentation and the workflow is given in there website you have to use the same as it is

I create a vm in the cloud ec2 instance.

a. Vpc
b. Subnet
c. Security group
d. Then only I create the vm

It will be created in default vpc

Terraform workflow

Wednesday, March 25, 2026 2:07 PM

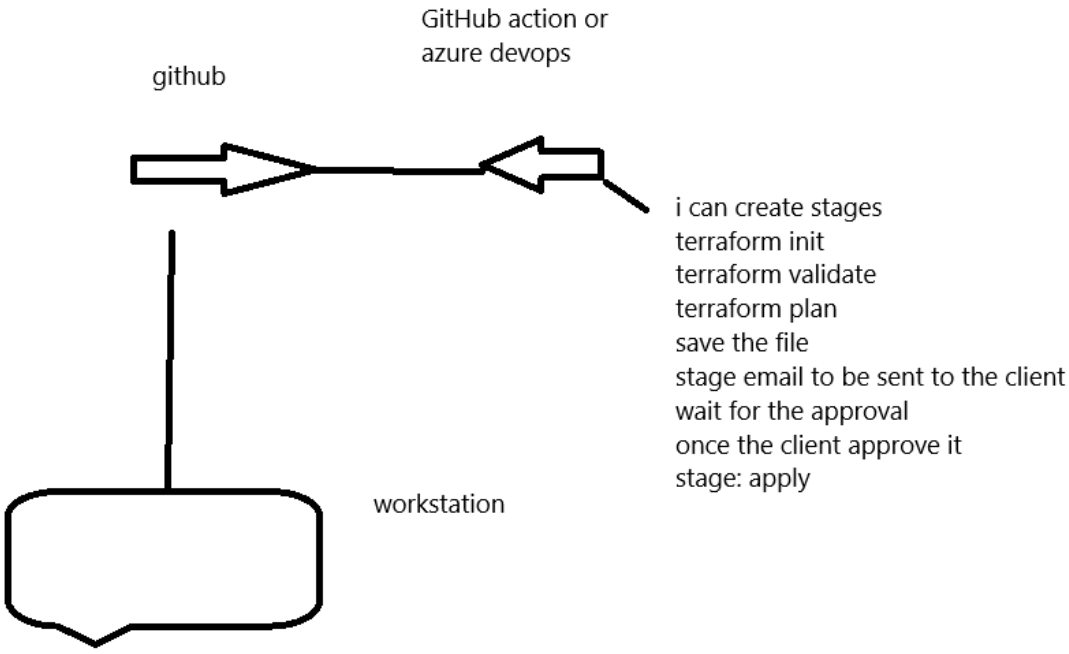
It is the standard process userd to safely develop test approve and deploy IAC. In real time secnarion this workflow also include security scanning policy approval and state management.

For eg. Like I got an request to create an ec2 instance using terraform.
Create a folder
Inside the folder we will put the code
We will start a teminal which and run our workflow.

Inside your project folder it will create a folder .terraform inside the same it will download the plugin required to communicate with the cloud service provider
It will convert your terraform code into an cloud provider speicific langauge. It do http request to the cloud service provider.

Terraform validate will not compare your code with the cloud service provider

Terraform plan: on the current directory it will read all your manifest file. Compare your code with the cloud service provider and provide an dry run or a blue print what it is going to create .,
You can use an --out commnad and store this blueprint inside a file also. ; we can put the same as an approval also



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Terraform apply: first it will scan your manifest file which contains resource create parameter. Inside that vpc subnet sg. Compare it with the cloud service provider.
Then create the resource for you in the cloud.
Convert your current state into desired state

Current state: which contains the code vpc subnet sg creation code
Desried state: it will create those resources in the cloud.

##once it created the resource in the cloud what next?
#it need to store those information inside a file called as terraform.tfstate

a. Audit
b. Idempotence

Store it in a remote and lock it

Terraform destroy

Wednesday, March 25, 2026 2:33 PM

I have manually create ec2 instance with RDS
On the next day I delete it also ec2 instance with rds
Elastics ip got created
Additional storage ideal
Vpc
21 month

Terraform destroy will run on the current directory it will read which file tfstate file and delete everything and make my state file blank.

Terraform apply t3 micro instance
T3.medium.
In the manifest vm configuration we will put t3.medium
Again it will read your all the manifest file compare it with tfsate file then generate a new plan
Vpc sg subnet ec2 instance type change
We need an storage which can do the versioing.
S3 enable versioing.

Drift
I have terraform code which contains vpc subnet sg ec2 instance
Vpc : 172.31.0.0/16
I am changing the cider block to 10.0.0.0/16 similar 10.0.1.0/24
Sg ec2 : in this type of secnario terraform apply
Everything will delete ec2 instance all the data will be deleted and a new instance will be created

Terraform resource block

Wednesday, March 25, 2026 2:55 PM

Using terraform what we want to create
Resource

I make change almost 10 days back
In my local machine I cannot do versioing.
So we will see how to store the state in a s3 azure blob storage
If I enable versioing and I upload the same file multiple times it is not going to overwrite it will create a version with timestamp
In local machine the current state is stored inside tfstate file and previous goes inside tfstate.backup file

Refrence block

Wednesday, March 25, 2026 4:37 PM

1. The first purpose of the refrence block is to store information against the refrence block inside your tfstate file
2. To map one resource with another resource

From this month end zomoto is going to increase the delivery fee

Recap

Thursday, March 26, 2026 9:38 AM

We learn about terraform workflow
Terraform init: it will create a . Terraform folder and as per the provider block it will download the cloud service provider plugin
Terraform validate: this will do just syntax validation
Terraform fmt: this will align your code
Terraform plan: dry run. This will run on existing folder current directory . Read all your manifest which has an extension of .tf compare the current state with desired and provide you an blue print. You can save the blue print using -out file name
Terraform apply: this will convert your current state into desired state it will store the state information inside a file state file
Terraform destroy: state file compare it with the cloud service provider and terminate all the resource

Terraform apply/terraform destroy --resource nameoftheresource specific resource

Variables

Thursday, March 26, 2026 10:02 AM

Input varaibles: there scope is global in nature
Output values block:

Once I create a ec2 instance or vpc: we go to console to verify the same amazon console

Terraform apply
It should give me the public ip or vpc id ec2 instance in the terraform console
Developer I don't want give access to the cloud service provider
Once the resource get created the public ip of the instance will be maill to him load balancer details

When you want to use name expression.
Local values is that local values have scope inside that folder only
#data source block
In aws which service I can use it to store my secret

secret manager
store my password

using terraform we are creating a windows machine.
when we create an windows machine
we need to provide password based authentication
inside my manifest file we have added the password
share the code with others they will also know the password
my password will be avaialbe inside my state file also

the purpose of the data block is that when we want to call anything outside terraform

but this ami is already avbaialbe in aws
i want to call this ami id

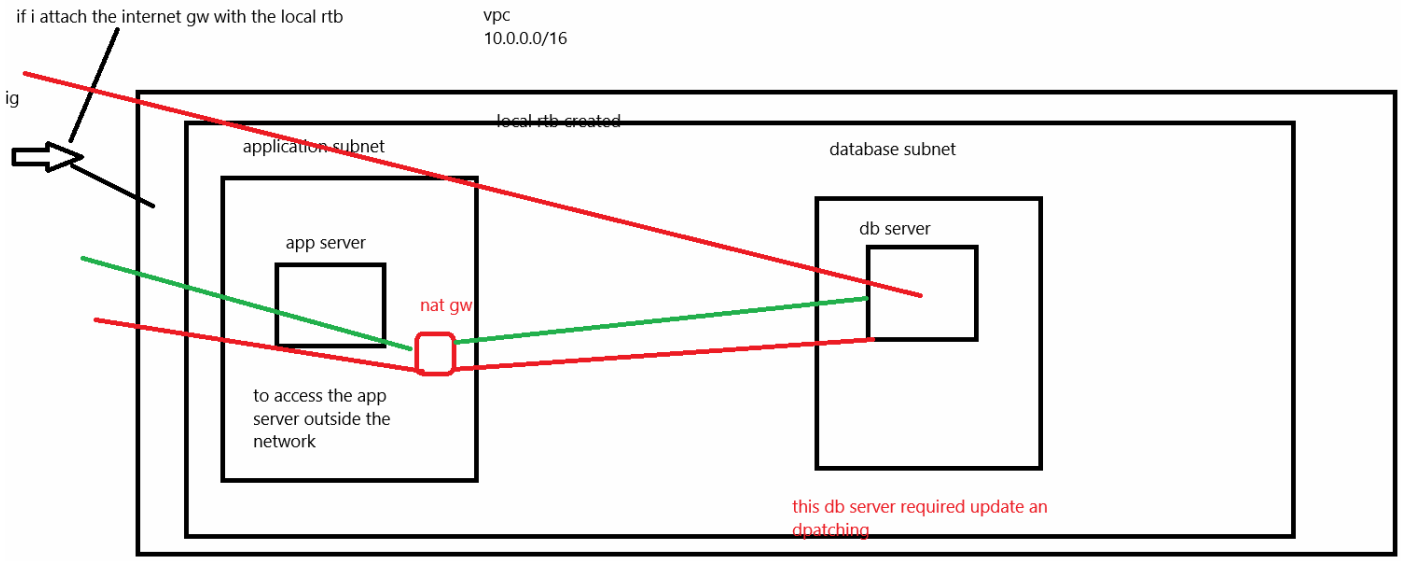
3/26/2026 10:14 AM - Screen Clipping

Demo

Thursday, March 26, 2026 10:18 AM

We will create an well secure architecture to deploy my resource in the cloud.

1. To create well secure architecture make your code readable using variables.
2. Not to hardcode any password



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vpc

Thursday, March 26, 2026 10:31 AM

This will be like an production style vpc

1. We will first create a varaibles file

Every resource block in terraform can create a single resource.
I need to create subnet multiple subnet
When we want to create multiple subnet simple way is to replicate the same block the code size increase and it is not an clean code
So what we can do it loops.
Terraform has a loop called as for_each which is similar to for loop

```
for (int i = 0; i < 5; i++) { for intitlzied,condtion,update
   System.out.println(i);
}
```

1. We have three subnet
2. And this three subnet need to be created in three az

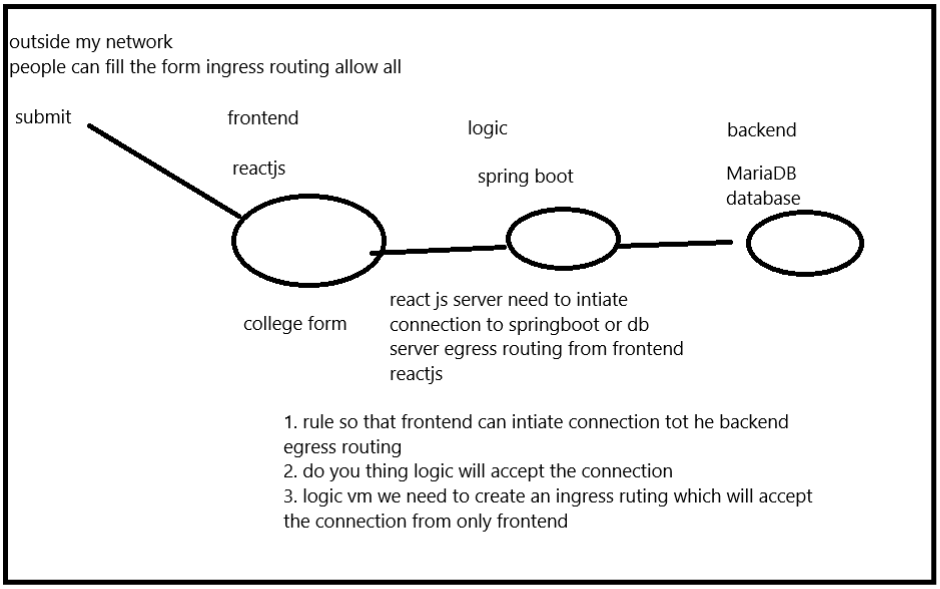
We want to create subnet
But with two condtion
Multiple subnet and spread across multiple region
We can use for_each this is a meta argument in terraform
It required intilization condtion and update

Any multiple values I want to display using output it should be value=values(resource_type.refrenceblock)[*].typeofresource like id public ip private ip
value = values(aws_subnet.sky_subnet)[*].id
##which this vpc we need to add an internet gw

Dynamic

Thursday, March 26, 2026 12:27 PM

For_each which is like a meta argument loop around a resource block
Now we have nested configuration.
For example in sg we want to allow incoming and outgoing ports multiple rules I need to create
Ingress and egress routing



disable everything

allow connection to frontend
2. acception intiate connection from frontend to logic
3. logic should accept the connection from frontend
ingress routing

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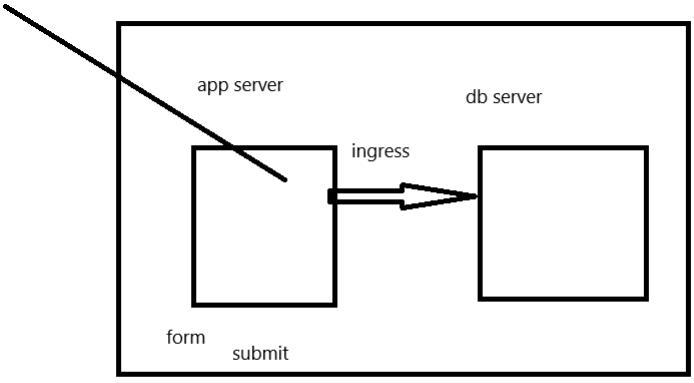


vpc
disable all the incoming and outgoing routing
1. first expetion i ll create where i need to allow connection to my frontend
outside the network ingress routing 0.0.0.0/0

2. app server want to save the data in db app-db-intiate-connection-ingress
routing
3. dbserver-appserver need to accept the connection egress

3/26/2026 12:44 PM - Screen Clipping

Dynamic or for_each

Thursday, March 26, 2026 2:14 PM

For_each
Required key and value menas when you want to play with key value pair
Suppose I want to create an instance load balancer

If I want to do looping around more than two values then I need to do nesting nesting using dynamic

Security group is stateful firewall: this means that if an inbound request is allowed the corresponding outbound will response automatically permiteed . We open all the ports
NACL is stateless

Ec2

Thursday, March 26, 2026 2:52 PM

Our networking
With subnet internet connection sg with desired port
Now we want to launch an ec2 instance
Which block I need to use to call an existing resource

Resource to create
Varaibles to create varaibles
Output block to create output
Data
Bootstrap script

Az

Thursday, March 26, 2026 4:35 PM

We have 4 azs currently on one single az one instance is being created.

Recap

Friday, March 27, 2026 9:41 AM

We learn about for_each loop in terraform.
Every resource block can create a single resource. But there are requirements like

1. Creating multiple subnet in multi az
2. Creating ec2 instance replication

The benefit for for each loop is that you can use the same block and create similar resource
Dynamic block when you want to loop around more than one single key.

Varaibles :
String varaible: single value
List of string: I can provide a list multiple subnet multiple az it should be in square bracket seprated by comma

Data block: anything which is created outside terraform ami I am launching an windows machine password

Map varaibles

Friday, March 27, 2026 9:48 AM

It is kind of an data types. Mainly used to assocaite key with a specific value especiually when you want to associate a key
Variables always required key and value
Key value data structure
It will store configuration data define varaibles and pass information between different parts of your code.

Dev test staging and prod environment
How to reduce the cost in this type of environment
Sizing of the instance
Billing in aws is too complex
I create an instance with 50Gb of hdd . We are paying for the instance and the storage.
If I shutdown the instance will storage cost will go away?
Shutdown the instance
Cpu ram comes into instance pricing if I shutdown the instance cpu ram pricing goes down
Storage
Dev: t2.micro
Test : t2.small
Prod=t3.medium

Terrform has something called as map(string)

Which is similar to dictionary in python.
Normally in dictionary against one single word

Named expression varaibles
Gopal : varaible
Expression should be always in key value format
Sad: wicked
Happy: smile

#there is an car company in eu
Instance_type

Dev = t2.micro--->t3.micro
10Gb of storage 12GB but in your os storage size increase will not be visible
Test = t2.small
Prod = t3.medium
#terraform apply

Remote state

Friday, March 27, 2026 10:58 AM

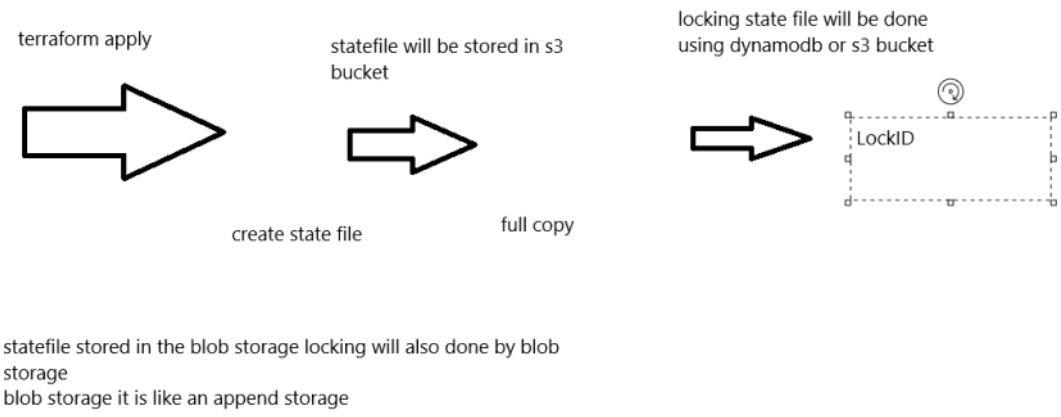
Currently we are storing the state file in my local machine.
The problem with my local machine hdd is that it cannot do the versioning.
Current state if stored in tfstate previous state goes to tfstate backup file.

In the past two days we have modify plenty of changes we have done it . We are loosing all the previous state file.

10 days back the customer told you to open port 22 they send an email you modify the terraform code and open the port 22
After few days banking complaince team comes to you that why did you open port 22
This resolve your issue
But the complance team whant to check for valunaribility
Can you provide the exact date and time when did you open the port

You can provide the state file
Why we are using this devops documenation is always created by the tool perfect documentation so we need not loose the document store it for a longer duration.
But you need an storage where if I upload the file multiple times. It should not be over written.
Amazon s3 has the capability where if you enable version the file will not be over written
Azure blob storage
Gcp storage
######you cannot use github or any repo to store your state file

We will create an s3 bucket
We will store the state file inside the s3 bucket and state locking

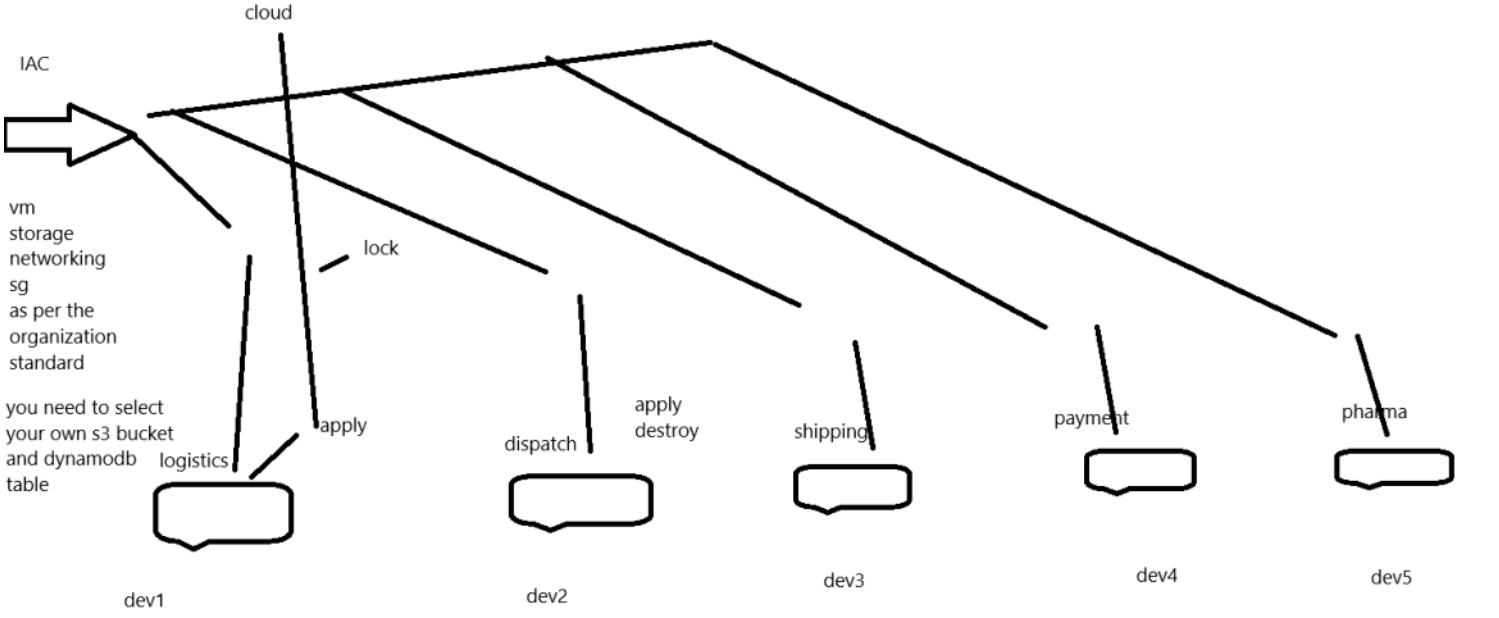


statefile stored in the blob storage locking will also done by blob storage
blob storage it is like an append storage

3/27/2026 11:45 AM - Screen Clipping

Never make your s3 bucket public
You need to also enable versioing in the s3 bucket

1. S3 bucket with versioing
2. Dynmodb table we will create with a row LockID



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ALB

Friday, March 27, 2026 12:32 PM

Application load balancer.
We are creating multiple instance and access those instance using individual public ip
In production we cannot tell the client
We need to provide an dns name using that traffic will be diverted to multiple vm.

1. First purpose of the load balancer is to divert traffic to multiple instance
2. We also required health checkup

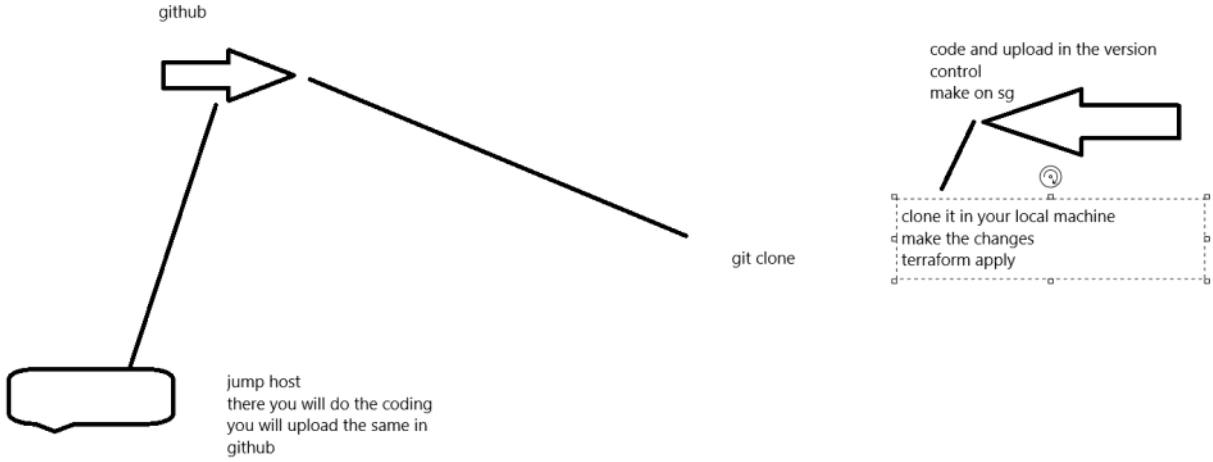
First we will create a new sg for my lb
Then create the load balancer
Attach all the subnet
Target group
Then attach all the instance
Listenenr which is going to send the traffic from lb to instance

Workflow in aws to create the alb amazon provided
You can download the code from terrform registry and build the resource

Alb-sg
Alb: how to attach this sg with alb refrence block

From one resource to another resource the mapping is done using refrence block

Lets do the hand on session on the same



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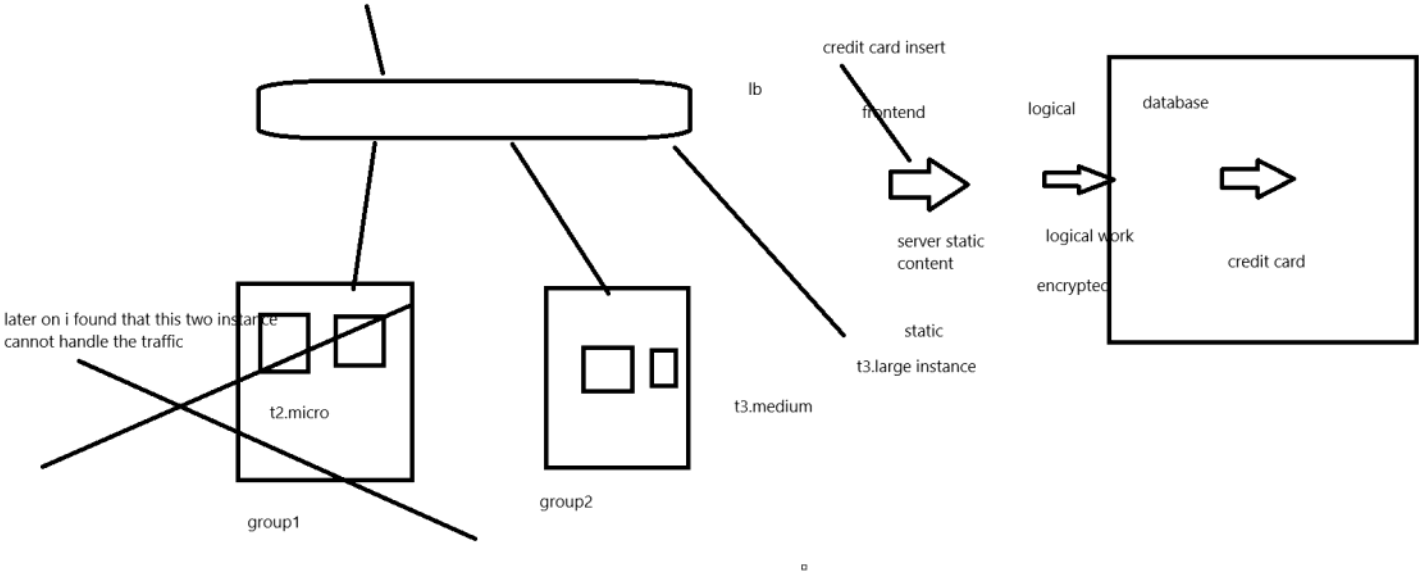
Inside target group we will attach all my instance
We will also do health checkup
Group 1 group2
Group 1 we will have instance with t2 micro in group2 we will have instance size as t3.micro
Group 1 : t3 m,icro but I don't want to delete the target group and recreate with new instance size
Group 2 t3 medium and attach it behind the load



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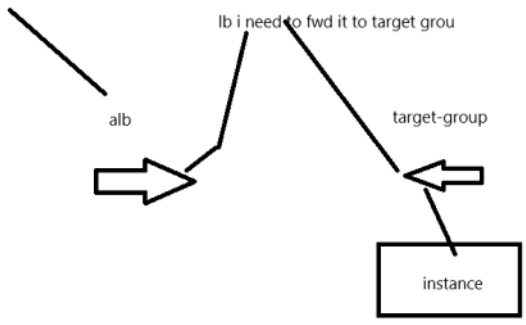
401 400 error http error
200 ok sucess



3/27/2026 3:04 PM - Screen Clipping

Finally we need to add the listener the target group is going to listen whom the load balancer or alb

Scaling

Friday, March 27, 2026 4:36 PM

Currently we have created the instance using manual scaling.
We want to do autoscaling
When instance get launched min 2 max 4
But min 2 we undestand how to increase another 2 instance
We need to provide an template
Configuration which containes the configuration for you ec2 instance.

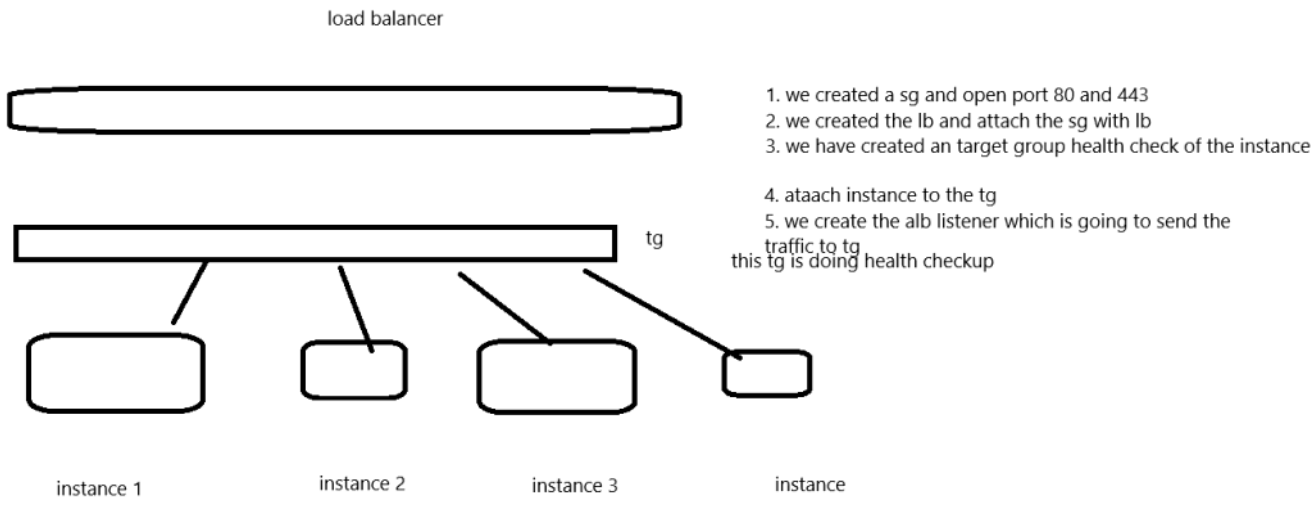
Autoscaling default behaviour
Min 4
And I have provided 4 subnet autoscaling will autmatically launch 4 instance in 4 different az

Recap

Saturday, March 28, 2026 9:43 AM

We started with a kind of varaibles called as map varaibles.
String varaible: it will accept one single default value
List of string will accept multiple values but in a list format
But I have a secanario where the default varibles need to pass in key value format. Similar to dictionary varaibles

Then we have created multiple instances. Using for each loop
We want to distribute the load to those instances behind your load balancer



3/28/2026 9:54 AM - Screen Clipping

Autoscaling

Saturday, March 28, 2026 9:55 AM

As the instance is created manually using terraform
The problem is it is costly if you have an monolothical architecture

But modern application build using micro services. We can use autoscaling

1. This will save my cost
2. In the peak time I can increase the number of instances manually

To create autoscaling we have created a template
Configuration file for instance
Min: 1 max 10
Using autoscaling we can create an ha kind of load

We are choosing az
Min: 2 max:4
Will it launch both the instance in same az. Where it will ensure that your workload is distributed across az
If the whole region goes down
Amazon is down n.verginai- application running snapchat till the n.vergi
Amazon prime down
Route 53

Autoscaling exercise

Saturday, March 28, 2026 10:12 AM

We have created autoscaling
Which create two instance min 2

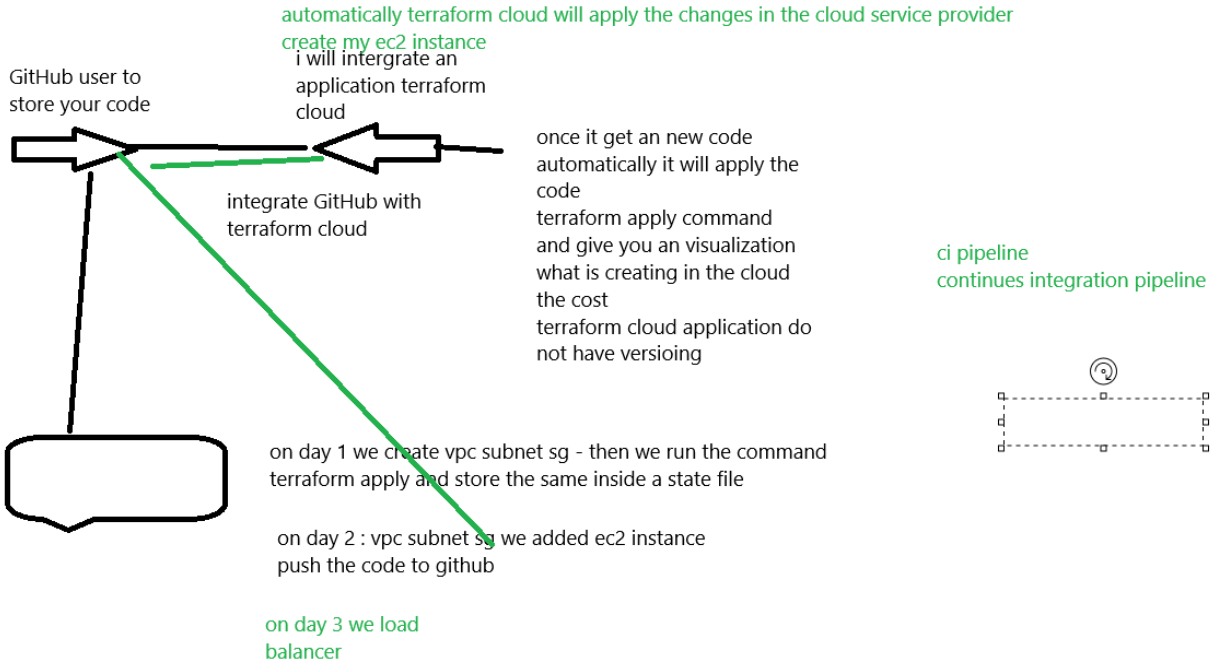
1. Add scaling policy cpu-scaling if my cpu percentage goes more than 50% in both the instance it should scale up add two more instance.
2. Apache bench tool we will use to do load testing
3. 10,000 request 200 concurent user

#lets reply the autoscaling

Git

Saturday, March 28, 2026 11:54 AM

Currently I am storing my secret inside local machine
But once I integrate with terraform cloud secret will be store in terraform cloud using vault



3/28/2026 12:08 PM - Screen Clipping

Continues devlier
I have created an well secure template
Sentinal policy

Creating a vpc : cidr block should be 11.0.0.0/16
Policy as code
Enforcing level:
The volume EBS should be encrypted
S3 bucket server side encryption
In security group port 22 and 3389 accessable using 0.0.0.0/0
Policy scope, advisory for ssh custom port number 2222 soft-mandatory hard-mandatory
Whenever a new terraform plan is run the connected workspace sentinal will automaticall evaluate the plan agains the policy . If you manditory policy fail the infra provison is blocked

1. Cost
2. Security

Ci pipeline
Cost
Visualization workflow
Security implement using sentinal

Pipeline

Saturday, March 28, 2026 12:39 PM

Learn about git
Git to manage your code
Git branching: using the same code I can create dev test staging environment
Then integrating git with terraform cloud

Git workflow

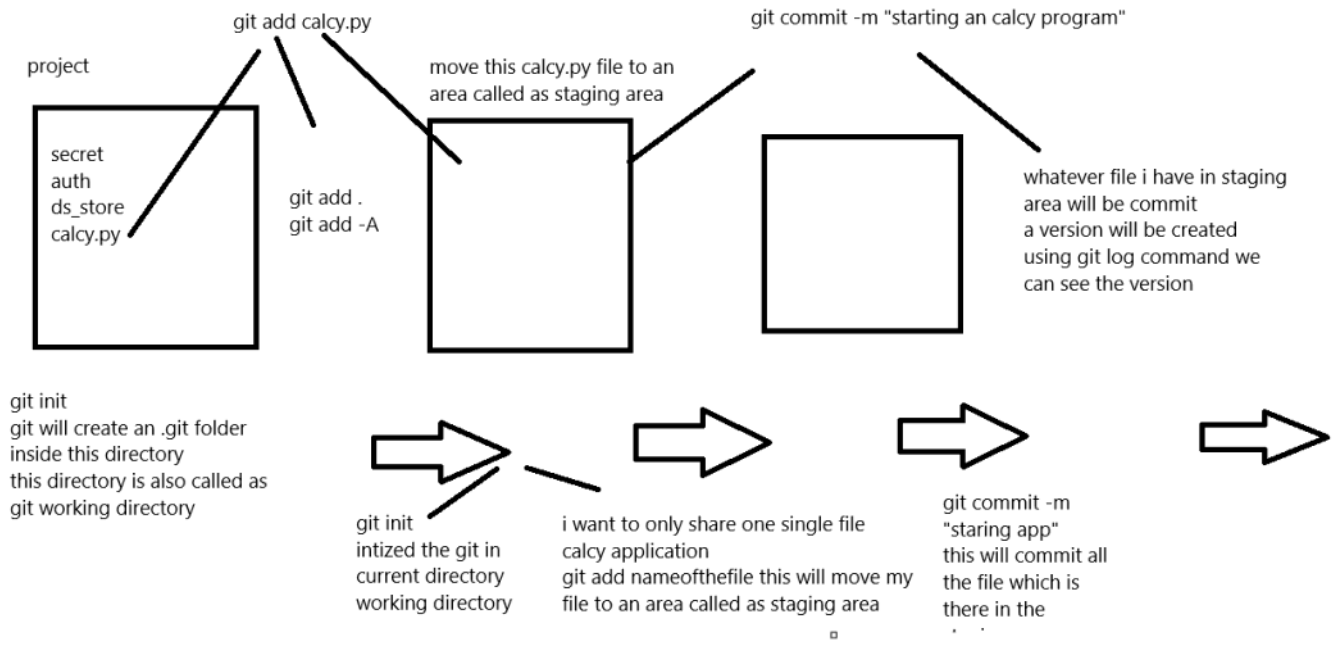
Saturday, March 28, 2026 2:05 PM

We will learn about git workflow
How using git we can manage the code
Whenever we start righting code IAC we need to manage the same using git

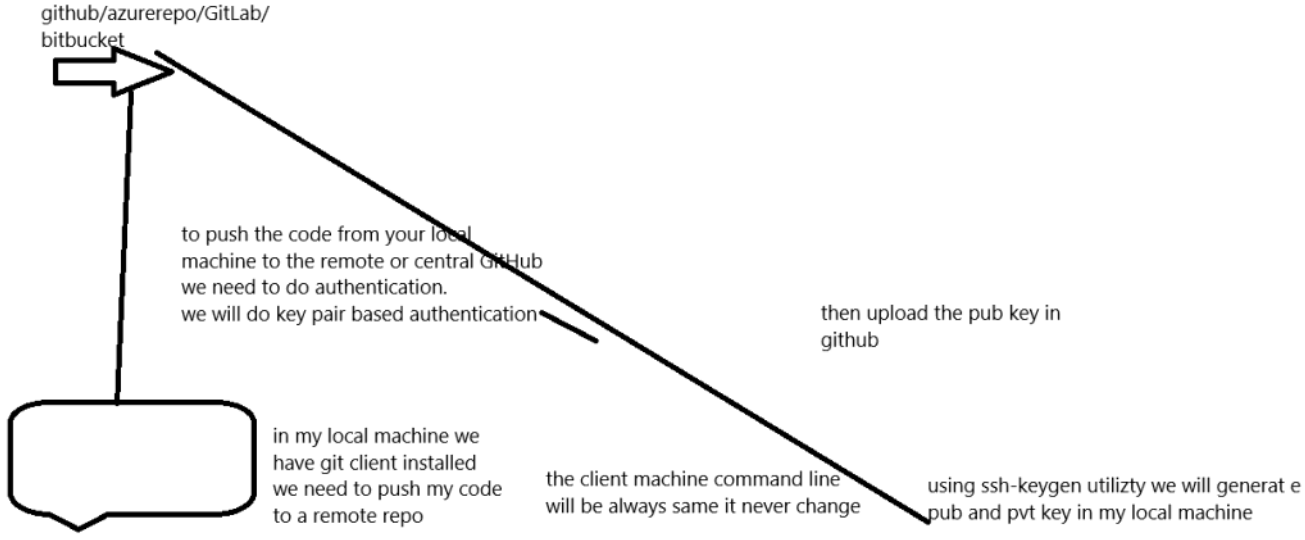
The first thing we will do we will learn about git global varaibles.
In global varaible we are creating username and email this is just for whenever you create a version automatically this global varaible will be attached with that version
I have create a folder inside the folder we created some files. Now I want to manage the folder using git
Think that the folder is like an workspace

We learn about git global varaible is that every time I commit the code the global varaible will be attached autmaticall with every commit who has done the commit the information are being stored

Git init in the current directory it will create a .git folder which is like an database for git to store all the versioing information.



3/28/2026 2:37 PM - Screen Clipping



3/28/2026 3:01 PM - Screen Clipping

Git branching

Saturday, March 28, 2026 3:34 PM

It will deverge from the main line of development and work on a new feature or bug fixes or isolation

ecommerce application

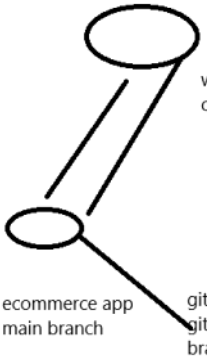
1. i dont want to disturb the skelton of application
2. i dont want to work from the scratch
3. just need to develop the new feature
4. but to develop the new feature i need that skelton

skeleton of my ecommerce application is develop running in production

i want to add a new feature platform fee

if i start building in the existing code and there will be some bug created

feature branch
platform fee



whatever code i have in main branch will be copied to the feature

from there i can develop this platfrom fee branch
git cheap copy mach

ecommerce app
main branch

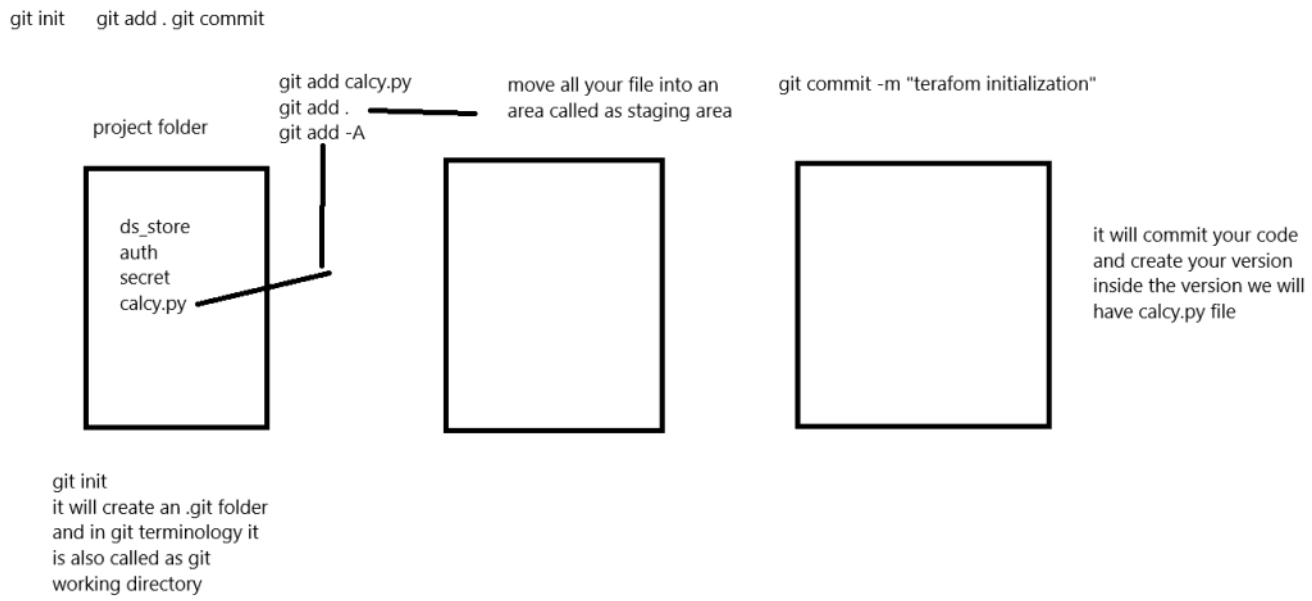
git branch feature
git checkout feature branch

3/28/2026 3:41 PM - Screen Clipping

Git

Sunday, March 29, 2026 9:43 AM

What is the git workflow



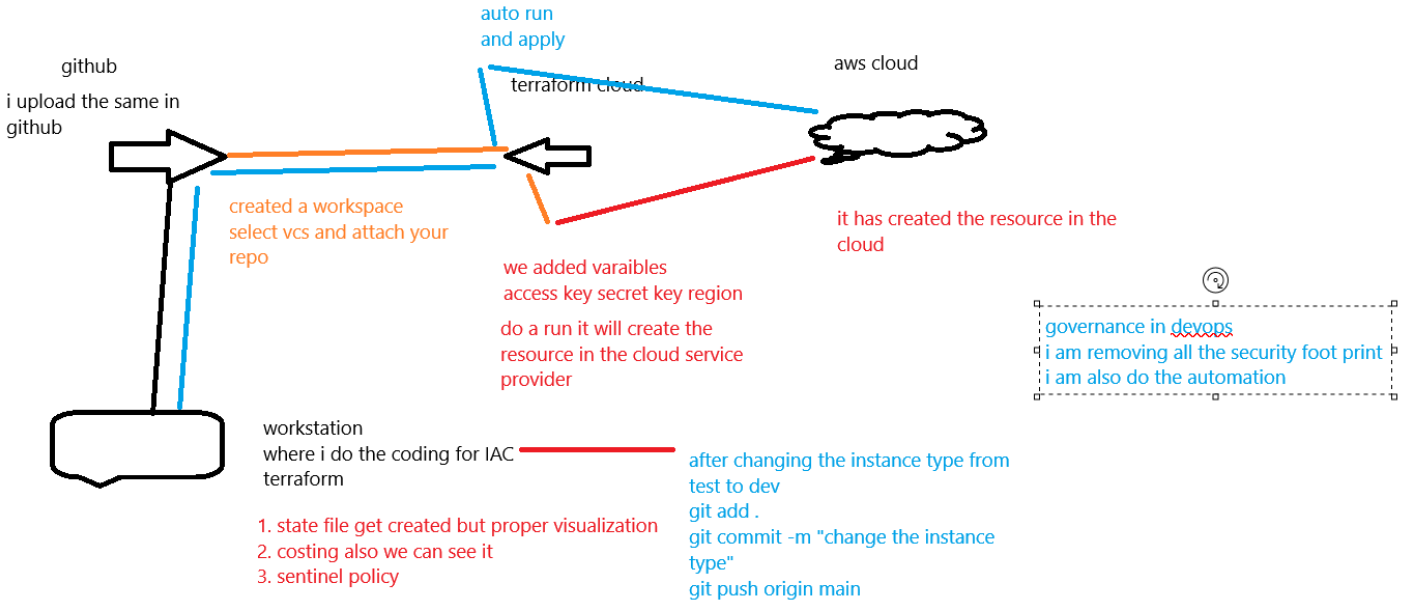
3/29/2026 9:47 AM - Screen Clipping

Every time I make changes in my code do we required git init to be done again and again one time

Git branching

Sunday, March 29, 2026 9:47 AM

Branches in the git
It will do an cheap copy withought disturbing my existing code
Dev test staing prod environment



3/29/2026 11:48 AM - Screen Clipping

Sentinel

Sunday, March 29, 2026 2:14 PM

What is this sentinel ?

1. Restrict instance size
2. Mandatory tags identify the resource and purpose
3. Block public ip
4. Required encryption. Ebs volume s3 bucket
5. Restric region. Flipkart business is specific to india.
6. Cost governance

Now to write down this policy we need to use python or you can use open policy agent opa : you need to know rego language.
So I got an policy that only t2.micro instance t2 small instance is allowed
How to apply the same we will see it inside the terraform cloud

Modules in terraform

Sunday, March 29, 2026 3:25 PM

What is this modules in terraform.